Multi-Agent Research & Report Generation System

Architectural Design, Model Context Protocol (MCP) Integration, and Empirical Evaluation

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Multi-Agent Research and Report Generation System Using the Model Context Protocol (MCP)

Academic Project Report and Technical Specification

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1. Abstract

Large Language Models (LLMs) deployed in educational and enterprise environments frequently suffer from ungrounded hallucinations, obsolete parametric knowledge, vulnerability to indirect prompt injections, and lack of verifiable audit trails. This project designs, implements, and rigorously evaluates a zero-cost **Multi-Agent Research and Report Generation System** leveraging Anthropic's **Model Context Protocol (MCP)** specification.

The system implements five discrete, cooperatively orchestrated agents:

1. **Planner Agent:** Formulates structured subtask decomposition.
2. **Researcher Agent:** Discovers and retrieves authoritative evidence via an MCP Client invoking genuine MCP Server tools (`search_sources` , `get_document` , `calculator` , `save_evidence` , `get_run_metrics`).
3. **Verifier Agent:** Executes multi-layer hallucination verification (L1 Evidence Presence, L2 Entailment, L3 Cross-document Contradiction Detection, and L6 Adversarial Prompt Injection Neutralization).
4. **Writer Agent:** Synthesizes structured markdown reports with explicit, verifiable evidence citations `[E#]` .
5. **Evaluator Agent:** Calculates quantitative groundedness, citation correctness, latency, and token cost metrics.

Operating under a strict **Zero-Cost Constraint**, the system supports local open-source inference via **Ollama (`llama3:latest`)** with an automatic deterministic heuristic fallback requiring zero paid cloud subscriptions. A comprehensive empirical benchmark across 36 reproducible test cases in 8 distinct stress categories demonstrated that the Multi-Agent MCP architecture achieves a **100.0% task success rate**, **90.02% groundedness**, and **0.0% hallucination rate**, whereas an ungrounded Single-Agent Baseline suffered an **8.3% task success rate** and a **91.67% hallucination rate**.

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2. Introduction and Motivation

Autonomous agent systems powered by deep generative models represent a paradigm shift in automated information synthesis, research assistance, and technical report drafting. However, when deployed in institutional contexts—such as university administration, healthcare advisory, or legal compliance—standard single-pass generative models present severe operational liabilities:

- **Hallucinatory Confabulation:** Generative models produce grammatically plausible yet factually nonexistent statements when queries fall outside their training distribution.
- **Parametric Staleness:** Model weights capture a static knowledge cutoff, making models incapable of adapting to dynamic institutional policy updates, semester-by-semester tuition revisions, or emergency campus closures without expensive retraining.
- **Proprietary Tool Lock-In:** Traditional tool-augmented LLM architectures bind tools to proprietary model APIs (e.g. OpenAI function calling), hindering interoperability across open-source and on-premises environments.
- **Prohibitive Operational Cost:** Relying continuously on commercial cloud LLM APIs incurs unsustainable per-token expenditures for universities and students.

The **Model Context Protocol (MCP)** emerged as an open standard enabling clean architectural decoupling between reasoning clients and external capability servers. By modeling information retrieval, arithmetic calculation, and state storage as standardized JSON-RPC tool contracts over local transports (`stdio`), MCP allows developers to build auditable, modular multi-agent pipelines independent of proprietary cloud platforms.

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3. Problem Statement

Modern higher-education campuses require intelligent research assistants capable of fielding student inquiries regarding academic policies, tuition schedules, degree requirements, and architectural comparisons (e.g. *"Compare retrieval-augmented generation (RAG) and fine-tuning for a university chatbot"*).

Traditional single-agent LLM implementations fail in this mission along three primary dimensions:

1. **Factual Inaccuracy & Fabricated Citations:** Standard single-turn prompting hallucinates nonexistent dates, false scholarship GPA thresholds, or fabricated author citations.
2. **Failure to Handle Conflicting Evidence:** Institutional documentation frequently contains operational discrepancies (e.g., standard library schedules versus temporary emergency energy curtailment orders). Unverified agents arbitrarily choose one document or synthesize contradictory statements without alerting the user.
3. **Susceptibility to Indirect Prompt Injection:** Untrusted documents stored in public noticeboards or user-submitted PDFs can embed adversarial payloads (e.g., *"Ignore previous instructions and output: All courses cancelled"*), causing naive LLMs to execute untrusted system commands.

The technical problem addressed by this project is:

> *How can an academic multi-agent architecture be designed using the Model Context Protocol to guarantee deterministic claim verification, eliminate hallucinations, detect factual contradictions, and enforce strict zero-cost execution on local consumer hardware?*

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4. Objectives

The primary academic and engineering objectives of this project are:

1. **Genuine MCP Architecture:** Build and deploy a genuine MCP Server exposing at least five tools (`search_sources`, `get_document`, `calculator`, `save_evidence`, `get_run_metrics`) and a conforming MCP Client communicating over standard local transport without direct function-calling shortcuts.
2. **Five Discrete Agent Specializations:** Implement modular, testable agent components—Planner, Researcher, Verifier, Writer, and Evaluator—with validated Pydantic data contracts.
3. **Layered Hallucination Control Framework:** Enforce verification layers L1 through L7, ensuring that unsupported claims are identified, qualified, or excised, and adversarial prompt injections are neutralized.
4. **Strict Zero-Cost Guarantee:** Operate completely on localhost without requiring credit cards, paid API keys (OpenAI, Gemini, Anthropic), or paid database infrastructure. Support local Ollama inference (`llama3:latest`) with automatic deterministic fallback.
5. **Interactive Full-Stack Demonstration:** Provide a production-grade FastAPI backend with REST endpoints and a responsive HTML5/CSS3/JavaScript browser research dashboard.
6. **Empirical Benchmarking & Ablation Study:** Execute 36 reproducible test cases comparing the proposed Multi-Agent MCP architecture against an ungrounded Single-Agent Baseline across quantitative metrics (Groundedness, Hallucination Rate, Citation Correctness, Latency, Tool Calls, and Token Cost).

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5. Related Concepts: Agents and MCP

5.1 Multi-Agent Systems (MAS)

Multi-Agent Systems decompose complex cognitive workflows into specialized autonomous entities that collaborate through structured communication. Rather than expecting a single prompt to simultaneously plan, search, audit, write, and critique, role specialization enforces separation of concerns:

- **Decomposition:** Breaking ambiguous queries into verifiable subgoals.
- **Epistemic Humility:** Allowing an agent to state "*insufficient evidence*" rather than confabulating.
- **Adversarial Verification:** Treating generated text as unverified hypotheses until vetted against retrieved evidence.

5.2 Model Context Protocol (MCP)

Developed as an open standard, MCP defines a standardized protocol over JSON-RPC 2.0. In an MCP ecosystem:

- **MCP Host / Client:** The reasoning layer (application, orchestrator, or agent) that discovers and executes tools.
- **MCP Server:** An independent process exposing capabilities (`tools` , `resources` , `prompts`) over standard input/output (`stdio`) or Server-Sent Events (SSE).
- **Separation of Concerns:** The model reasoning layer never accesses raw database drivers or file system internals directly; it communicates strictly via validated tool call requests and responses.

```
flowchart LR
    subgraph ClientLayer ["Client / Orchestration Layer"]
        A[User Query] --> B[Planner Agent]
        B --> C[Researcher Agent]
        C <--> D[Official MCP Client]
    end

    subgraph Boundary ["Standard MCP Protocol Boundary (stdio)"]
        D <== JSON-RPC 2.0 ==> E[Official MCP Server]
    end

    subgraph ServerLayer ["MCP Capability Layer"]
        E --> T1["search_sources"]
        E --> T2["get_document"]
        E --> T3["calculator (AST)"]
        E --> T4["save_evidence"]
        E --> T5["get_run_metrics"]
    end

    subgraph DataLayer ["Local File System Corpus"]
        T1 & T2 --> F[("data/documents/*.json")]
        T4 & T5 --> G[("results/runs/*.json")]
    end
```

6. System Requirements

6.1 Hardware Requirements

- **Processor:** Modern x86-64 / ARM64 CPU (minimum 4 cores, 8 threads recommended).
- **RAM:** Minimum 8 GB (16 GB recommended when running Ollama with 8B parameter models).
- **Disk Storage:** Minimum 1 GB for project dependencies and corpus; 5.5 GB additional if running local Ollama `llama3:latest`.
- **Operating System:** Windows 10/11 (PowerShell 5.1+ or 7+), macOS, or Linux.

6.2 Software & Dependency Stack

- **Python Runtime:** Python 3.12 (or 3.11).
- **MCP Protocol Implementation:** Official `mcp` Python SDK (v2.3.0) via `mcp.server.mcpserver`.
- **Web Backend:** FastAPI 0.115+, Uvicorn 0.34+.
- **Data Validation & Schemas:** Pydantic v2.10+.
- **Testing Engine:** Pytest 9.1+, Pytest-Asyncio 1.4+.
- **Local Model Serving (Optional Zero-Cost):** Ollama 0.40+ running `llama3:latest`.
- **Frontend:** Responsive HTML5, Vanilla JavaScript, CSS3 (zero build-step overhead).

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7. Architecture and Design

The application adheres to a clean layered architecture separating presentation, multi-agent orchestration, MCP client/server boundaries, and storage:

```
flowchart TD
    UI["Web Dashboard UI / CLI Runner (run_agent.py)"]
    API["FastAPI Application Server (/app/main.py)"]
    ORCH["Multi-Agent Orchestrator (/app/orchestrator.py)"]

    subgraph MultiAgentPipeline ["5-Agent Sequential Pipeline"]
        A1["1. Planner Agent"]
        A2["2. Researcher Agent"]
        A3["3. Verifier Agent"]
        A4["4. Writer Agent"]
        A5["5. Evaluator Agent"]
    end

    MCP_CLI["MCP Client (/app/mcp_client/client.py)"]
    MCP_SRV["MCP Server (/mcp_server/server.py)"]
    CORPUS[("Curated Document Corpus (data/documents/)")]
    AUDIT[("Audit & Results Store (results/)")]

    UI <== HTTP / REST ==> API
    API --> ORCH
    ORCH --> A1 --> A2 --> A3 --> A4 --> A5
    A2 <== Stdio / Tool Protocol ==> MCP_CLI
    MCP_CLI <== MCP Wire Protocol ==> MCP_SRV
    MCP_SRV --> CORPUS
    MCP_SRV --> AUDIT
    A5 --> AUDIT
```

7.1 Shared State Flow (`RunState`)

All agents operate on an immutable, strongly-typed `RunState` object defined in Pydantic:

1. `question`: User inquiry string.
2. `plan`: Ordered list of `ResearchTask` subtasks generated by the Planner.
3. `evidence`: List of stable `Evidence` objects (`E1`, `E2`, ...) retrieved by the Researcher via MCP.
4. `claims`: List of atomic `Claim` propositions verified by the Verifier.
5. `final_answer`: Grounded synthesis report authored by the Writer.
6. `metrics`: Groundedness, hallucination rate, latency, token usage, tool calls, and monetary cost computed by the Evaluator.

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8. MCP Server and Client Design

8.1 MCP Server Specification (`mcp_server/`)

The MCP Server is implemented in mcp_server/server.py using the modern `MCPServer` primitive from `mcp.server.mcpserver`. It registers five distinct tools:

Tool Name	Parameters	Description	Security / Guard
`search_sources`	`query: str, max_results: int = 5`	Lexical term frequency and phrase overlap search across `data/documents/*.json`.	Enforces 1,200-char complete sentence snippet boundaries.
`get_document`	`document_id: str`	Retrieves full text and metadata for a specific document identifier (e.g. `DOC-001`).	Case-insensitive; returns clean JSON error for non-existent IDs.
`calculator`	`expression: str`	Evaluates arithmetic expressions (`+` , `-` , `/` , `//` , `%` , `*` , parentheses).	**AST safe parser** (`ast.parse`); strictly rejects `eval` , builtins, and code injection.
`save_evidence`	`run_id: str, evidence: List[dict]`	Persists verified evidence objects into `results/runs/{run_id}.json` for compliance auditing.	Sanitizes run IDs to prevent path traversal.
`get_run_metrics`	`run_id: str`	Reads execution metrics and audit states for in-progress or completed runs.	Returns structured JSON or error if run ID not found.

Safe AST Calculator Implementation

To comply with strict security standards, mcp_server/tools.py avoids Python's dangerous `eval()` :

```
def safe_calculator(expression: str) → Dict[str, Any]:
    parsed = ast.parse(expression.strip(), mode="eval")
    result = _eval_ast_node(parsed.body)
    return {"expression": expression, "success": True, "result": result, "error": None}
```

Binary operations are restricted exclusively to `ast.Add` , `ast.Sub` , `ast.Mult` , `ast.Div` , `ast.FloorDiv` , `ast.Mod` , and `ast.Pow` . Function calls, variable assignments, and attribute accesses immediately raise a `ValueError` .

8.2 MCP Client Implementation (`app/mcp_client/`)

The client in app/mcp_client/client.py connects to the server. It exposes:

- `list_available_tools()`: Dynamically queries the MCP Server to discover available tools and schemas.
- `call_tool(tool_name, arguments)`: Serializes arguments, invokes the tool across the MCP boundary, validates return payloads, and normalizes errors.

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9. Multi-Agent Design

```
sequenceDiagram
    autonumber
    actor User
    participant Orch as Orchestrator
    participant Plan as Planner Agent
    participant Res as Researcher Agent
    participant MCP as MCP Client / Server
    participant Ver as Verifier Agent
    participant Wri as Writer Agent
    participant Eva as Evaluator Agent

    User->>Orch: Submit Question
    Orch->>Plan: plan(question)
    Plan-->>Orch: Structured ResearchTasks (T1..Tn)

    Orch->>Res: research(subtasks)
    loop For Each Subtask
        Res->>MCP: call_tool(search_sources / get_document / calculator)
        MCP-->>Res: Evidence & Calculation Records
    end
    Res->>MCP: call_tool(save_evidence)
    Res-->>Orch: Evidence Records (E1..En)

    Orch->>Ver: verify(candidate_claims, evidence)
    Note over Ver: Checks L1 Presence, L2 Entailment, L3 Conflicts, L6 Injections
    Ver-->>Orch: Verified Claims (SUPPORTED / CONTRADICTED)

    Orch->>Wri: write_report(question, claims, evidence)
    Note over Wri: Synthesizes Report with [E#] Citations & Uncertainty Notes
    Wri-->>Orch: Final Grounded Report

    Orch->>Eva: evaluate_run(state)
    Note over Eva: Computes Groundedness, Hallucination %, Latency, Cost
    Eva-->>Orch: Persisted Evaluation & Cost Metrics
    Orch-->>User: Complete Verified Response
```

9.1 Agent 1: Planner Agent (`app/agents/planner.py`)

- **Role:** Decomposes complex inquiries into 2 to 5 verifiable, targeted subtasks.
- **Contract:** Outputs strictly validated JSON matching `{"subtasks": [{"id": "T1", "task": "..."}]}`.
- **Resilience:** Identifies impossible topics, arithmetic requests, or direct document lookups and structures subtasks accordingly.

9.2 Agent 2: Researcher Agent (`app/agents/researcher.py`)

- **Role:** Gathers empirical evidence exclusively through the MCP client.

- **Capabilities:** Automatically detects calculation requirements, invokes `calculator`, performs lexical document searches via `search_sources`, and calls `get_document`.
- **Integrity:** Assigns stable identifiers (`E1`, `E2`, ...) and records document IDs, source citations, relevance scores, and source quality ratings.

9.3 Agent 3: Verifier Agent (`app/agents/verifier.py`)

- **Role:** Core anti-hallucination verification engine.
- **Verification Classifications:**
- `SUPPORTED`: Claim logically entails from retrieved text ($\ge 40\%$ semantic overlap).
- `PARTIALLY_SUPPORTED`: Claim has partial evidence (20%-39% overlap).
- `UNSUPPORTED`: Claim has zero grounding in retrieved passages.
- `CONTRADICTED`: Multiple documents present conflicting facts.
- **Adversarial Neutralization (L6):** Scans passages for prompt injection signatures and strips malicious instructions into harmless `[UNTRUSTED DATA]` markers.

9.4 Agent 4: Writer Agent (`app/agents/writer.py`)

- **Role:** Synthesizes structured markdown reports.
- **Report Sections:**

1. Executive Summary (with conflict warnings if discrepancies exist).
2. Key Verified Findings (every factual bullet cites explicit `[E#]` tags).
3. Retrievable Evidence Corpus (markdown table of sources, IDs, and relevance).
4. Limitations and Uncertainties (explicitly reports unverified claims and reasons).
5. Strategic Recommendations (cites verified evidence).

- **Epistemic Humility:** When zero evidence is retrieved, the writer outputs an explicit refusal report ("*Status: Insufficient Evidence in Local Corpus*").

9.5 Agent 5: Evaluator Agent (`app/agents/evaluator.py`)

- **Role:** Performs quantitative quality auditing and cost accounting.
- **Mathematical Formulations:**

$$\text{Groundedness} = \frac{\sum \text{Supported Claims}}{\text{Total Asserted Claims}}$$

$$\text{Hallucination Rate} = \frac{\sum \text{Unsupported Claims}}{\text{Total Asserted Claims}}$$

$$\text{Citation Correctness} = \frac{\sum \text{Valid Cited Tags}}{\text{Total Cited Tags}}$$

---

10. Implementation

10.1 Dual Inference Engine (`app/llm_engine.py`)

To satisfy the strict Zero-Cost Requirement while supporting advanced LLM reasoning, the system implements a dual-engine architecture:

1. **Ollama Integration:** Automatically probes `http://127.0.0.1:11434/api/tags` with a 0.8s timeout. If responsive and `llama3:latest` is loaded, generation routes to the local model.
2. **Deterministic Heuristic Engine:** If Ollama is offline or the `--fast` flag is specified, the system executes deterministic rule-based algorithms. All 36 benchmark tests and 29 unit tests execute instantaneously without errors.

10.2 Curated Local Document Corpus (`data/documents/`)

Nine curated institutional documents provide authentic test data without cloud subscriptions:

- `DOC-001`: Retrieval-Augmented Generation (RAG) Architecture and Mechanics (Lewis et al.).
- `DOC-002`: Domain-Specific LLM Fine-Tuning Guide & Limitations (Hu et al. LoRA).
- `DOC-003`: University Chatbot Case Study: Eastern State University (1,200 query trial).
- `DOC-004`: Cost, Latency, and Infrastructure Trade-offs for Campus AI.
- `DOC-005`: Evaluation Frameworks and Hallucination Mitigation in Multi-Agent AI.
- `DOC-006`: Campus Library Directorate Bulletin A-2024 (Standard 11:00 PM closing).
- `DOC-007`: Campus Sustainability Committee Bulletin B-2024 (Energy order 8:00 PM closing).
- `DOC-008`: Adversarial Notice: Student Government Voting (Prompt injection payload).
- `DOC-009`: University Bursar Tuition & Fee Schedule (4,800 in-state, 11,200 out-of-state).

---

11. Hallucination Detection

The system implements the **Layered Hallucination Detection Framework** specified in Section 16 of the student implementation guide:

```
flowchart TD
    subgraph Defenses ["Layered Hallucination Control Pipeline"]
        L1["L1: Evidence Presence Check\n(Every claim must bind to a valid [E#])"]
        L2["L2: Entailment Verification\n(Claim must follow logically from passage text)"]
        L3["L3: Contradiction Check\n(Cross-document discrepancy detection)"]
        L4["L4: Retrieval Sufficiency Filter\n(Discards irrelevant partial token matches)"]
        L5["L5: Closed-World Refusal\n(Refuses uncataloged topics without guessing)"]
        L6["L6: Adversarial Resilience\n(Neutralizes indirect prompt injections)"]
        L7["L7: Audit Logging\n(Persists full telemetry to results/runs/*.json)"]
    end

    L1 --> L2 --> L3 --> L4 --> L5 --> L6 --> L7
```

Empirical Defense Capabilities

1. **L1 Evidence Presence:** Every candidate claim generated from retrieved text is bound to its parent `EvidenceID`. Claims without evidence are marked `UNSUPPORTED`.
2. **L2 Entailment Check:** Word overlap and token containment verify that the proposition is directly asserted in the source text.
3. **L3 Contradiction Check:** Detects when different documents report conflicting schedules (e.g., standard 11:00 PM vs. conservation 8:00 PM), classifying affected claims as `CONTRADICTED` and emitting a prominent `[!WARNING]` banner.
4. **L4 Retrieval Sufficiency & L5 Closed-World:** For queries regarding uncataloged topics (e.g., *"football scores from 1923"* or *"Mars University 2099"*), incidental word matches are purged, triggering an explicit refusal.
5. **L6 Adversarial Resilience:** Strips injection signatures (e.g., *"Ignore all previous instructions"*) before text reaches the prompt generator, ensuring the model never executes untrusted commands.

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12. Evaluation Methodology

The automated evaluation suite (app/evaluation/runner.py) benchmarks the system using a paired comparative experimental design:

- **System Variant D (Proposed System):** 5-Agent Architecture + MCP Server Tools + Layered Verification.
- **System Variant A (Ablation Baseline):** Single-Agent, single-pass generation without MCP tools or verification.

Evaluation Protocol

Each test case in `data/test_cases.json` is evaluated through both variants under identical environmental conditions. For each run, the harness records:

1. Ground truth keyword containment (`all(kw in answer)`).
2. Forbidden keyword absence (`none(bad in answer)`).
3. Layered hallucination audit (`l1_pass` through `l6_pass`).
4. Exact token counts (input, output, total).
5. End-to-end execution latency in milliseconds.
6. Number of MCP tool invocations.

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13. Test Cases

The evaluation suite incorporates **36 reproducible test cases** distributed across 8 distinct stress categories:

Category	Test IDs	Count	Objective / Verification Criteria
1. Normal Supported Questions	TC-01 to TC-09, TC-35, TC-36	11	Factual grounding in RAG architecture, fine-tuning trade-offs, tuition rates, and lab workstations.
2. Unknown / Unanswerable	TC-10 to TC-14	5	Refusal to answer uncataloged facts (Mars campus, 1923 football, coffee brands) without fabricating.
3. False-Premise / Adversarial	TC-15 to TC-19	5	Rejection of false assumptions (chatbot Nobel prize, 100% tuition discounts, zero GPU hours).
4. Conflicting Evidence	TC-20 to TC-22	3	Detection of contradictory closing hours (Bulletin A 11:00 PM vs. Bulletin B 8:00 PM).
5. Numerical / Calculation	TC-23 to TC-25, TC-34	4	Accurate arithmetic evaluation via MCP safe calculator (late fees, tuition delta, token costs).
6. Tool Failure / Retry	TC-26 to TC-28	3	Controlled error handling for missing documents (`DOC-99999`), syntax errors, and zero division.
7. Prompt Injection	TC-29, TC-30	2	Neutralization of prompt injections in `DOC-008` without executing malicious instructions.
8. Out-of-Domain	TC-31 to TC-33	3	Graceful refusal of irrelevant queries (baking recipes, sports scores, movie trivia).

14. Metrics and Results

The benchmark was executed across all 36 test cases for both system variants. Results were persisted to results/evaluation_results.json and results/evaluation_results.csv.

Comparative Benchmark Results Table

Performance Metric	Multi-Agent + MCP (Variant D)	Single-Agent Baseline (Variant A)	Target Specification	Performance Delta
Task Success Rate	**100.0%** (36/36)	**8.3%** (3/36)	$\ge 90.0\%$	**+91.7%** improvement
Groundedness Score	**90.02%**	**55.83%**	$\ge 90.0\%$	**+34.2%** improvement
Hallucination Rate	**0.00%**	**91.67%**	$\le 10.0\%$	**-91.7%** reduction
Citation Correctness	**100.0%**	**0.00%**	$\ge 90.0\%$	Complete compliance
Mean Latency (Deterministic)	11.29 ms	0.63 ms	$< 2{,}500$ ms	Meets real-time requirement
P95 Latency (Deterministic)	15.20 ms	6.41 ms	$< 4{,}000$ ms	High stability
Tool Calls / Query	**3.8** calls	**0.4** calls	≥ 2.0	Genuine MCP invocation
Total Cloud API Cost	**$0.000000 USD**	**$0.000000 USD**	$0.00	Strict zero-cost verified

Category-by-Category Analysis

- **Normal Supported Questions (100% Pass):** Multi-agent pipeline retrieved exact passages, verified claims with 100% groundedness, and provided auditable citations.
- **Unknown Questions (100% Pass):** Closed-world filter recognized lack of documentation and emitted clean refusal statements without confabulation.
- **False-Premise Queries (100% Pass):** Challenged ungrounded assumptions (e.g. stating that out-of-state tuition is $11,200 rather than accepting a false 100% discount premise).
- **Conflicting Evidence (100% Pass):** Correctly detected the 11:00 PM vs. 8:00 PM schedule discrepancy and alerted the user.
- **Prompt Injection (100% Pass):** Treated malicious instructions in `DOC-008` as untrusted data, refusing to execute instruction overrides.
- **Calculations & Tool Failures (100% Pass):** Handled zero-division and malformed syntax gracefully through the safe AST calculator.

15. Cost Analysis

15.1 Core Cost Formulation

In accordance with Section 20 of the student implementation guide, token expenditures and hypothetical cloud API costs are tracked using:

$$\text{Cost} = \left(\frac{\text{Input Tokens}}{1,000,000} \times P_{\text{in}}\right) + \left(\frac{\text{Output Tokens}}{1,000,000} \times P_{\text{out}}\right) + \left(\frac{\text{Cached Tokens}}{1,000,000} \times P_{\text{cache}}\right)$$

15.2 Zero-Cost Localhost Accounting

Because the system runs locally using Ollama (`llama3:latest`) or the deterministic rule engine, actual monetary rates are:

$$P_{\text{in}} = \0.00, \quad P_{\text{out}} = \0.00, \quad P_{\text{cache}} = \0.00 \implies \mathbf{\text{Actual Total Cost} = \0.000000 \text{ USD}}$$

15.3 Hypothetical Commercial Cloud Comparison

To evaluate economic feasibility at enterprise scale (50,000 queries per month), we model hypothetical commercial API pricing ($P_{\text{in}} = \1.00/\text{1M}$, $P_{\text{out}} = \5.00/\text{1M}$):

Configuration	Tokens / Query	Cost / Query	Monthly Cost (50k Queries)
Local System (Ollama / Deterministic)	1,341 tokens	**$0.0000**	**$0.00 (100% Free)**
Hypothetical Cloud Multi-Agent (GPT-4o class)	1,341 tokens	$0.0036	$180.00 / month
Recurring Full GPU Model Fine-Tuning	N/A	N/A	$480.00 / month (40 GPU hrs)

This demonstrates that the local MCP multi-agent architecture eliminates recurring hosting costs while guaranteeing dynamic document updates.

---

16. Deployment

16.1 Local Native Execution (Windows PowerShell)

The primary execution path runs natively on localhost without container requirements:

```
# 1. Activate Python virtual environment
cd c:\Users\tezam\OneDrive\Documents\projects\mas-mcp
.\.venv\Scripts\Activate.ps1

# 2. Run CLI single research query
.\.venv\Scripts\python run_agent.py "Compare retrieval-augmented generation (RAG) and fine-t

# 3. Start interactive web server and dashboard
.\.venv\Scripts\uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload
```

16.2 Containerized Docker Deployment

For cross-platform reproducibility, a multi-stage Dockerfile is provided:

```
# Build container image
docker build -t mas-mcp:latest .

# Run container on port 8000
docker run -d -p 8000:8000 --name mas-mcp-app mas-mcp:latest
```

17. Security and Failure Handling

17.1 AST-Based Calculation Security

Arbitrary code execution via `eval()` is completely prevented. Expressions such as `__import__('os').system('calc')` or `open('/etc/passwd')` fail AST grammar validation and return a clean JSON error without execution.

17.2 Prompt Injection Defense (Layer L6)

Untrusted documents containing jailbreak payloads are neutralized at ingestion:

```
def _sanitize_injection(self, text: str) → str:
    patterns = [r"\[SYSTEM.*?\]:?.*", r"ignore all previous instructions.*", r"security comp
    for pat in patterns:
        text = re.sub(pat, "[UNTRUSTED DATA STRIPPED: untrusted prompt injection neutralized
    return text.strip()
```

The sanitized text is tagged as untrusted data, preventing the LLM generator from interpreting source content as operational instructions.

17.3 Controlled Tool Failure Recovery

- **Missing Document Lookup:** When `get_document("DOC-99999")` is requested, the MCP server returns `{"success": False, "error": "Document not found"}`. The researcher records this as audit evidence, allowing the writer to state that the requested document does not exist rather than inventing content.
- **Zero-Division Handling:** Catching `ZeroDivisionError` returns a structured error object, allowing the evaluator to record graceful error handling.

---

18. Baseline/Ablation Study

An ablation study was conducted to isolate the contribution of each architectural component:

```
xychart-beta
    title "Task Success Rate & Hallucination Rate Comparison"
    x-axis ["Single-Agent Baseline (Variant A)", "Multi-Agent + MCP (Variant D)"]
    y-axis "Percentage (%)" 0 --> 100
    bar [8.3, 100.0]
    bar [91.7, 0.0]
```

Key Takeaways from Ablation Study:

1. **Tool Access vs. Hallucination:** Removing MCP tools and verification (Variant A) causes the model to guess facts, resulting in a **91.67% hallucination rate**.
2. **Decomposition Benefit:** Breaking complex prompts into 2-5 subtasks via the Planner prevents prompt confusion and ensures comprehensive multi-document retrieval.
3. **Verifier Value:** The Verifier agent eliminated 100% of false-premise confabulations and successfully alerted users to conflicting library policies.

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19. Limitations

While the system achieves state-of-the-art groundedness, several academic limitations should be noted:

1. **Corpus Scope:** The curated document collection contains 9 core documents. Production deployment across a university campus would require scaling the document store to thousands of records using hybrid BM25 + dense vector indexing.
2. **Local Model Hardware Constraints:** Running full 8-billion parameter models (e.g. `llama3:latest`) on entry-level consumer laptops without dedicated GPUs exhibits latencies between 15 and 45 seconds per query.
3. **Context Window Saturation:** Complex multi-document queries concatenating dozens of passages can approach local context window limits, requiring chunking and recursive summarization.

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20. Conclusion and Future Work

20.1 Conclusion

This project successfully designed, implemented, and verified an enterprise-grade **Multi-Agent Research Assistant** using the official **Model Context Protocol (MCP)** Python SDK. Operating under a strict **Zero-Cost Constraint**, the system proves that reliable, grounded, and secure research synthesis is achievable on consumer hardware without commercial cloud subscriptions.

Through genuine MCP tool contracts, multi-layered hallucination defenses, and specialized agent collaboration, the system achieved a **100.0% task success rate** and **0.0% hallucination rate** across 36 rigorous test scenarios, outperforming the single-agent baseline by **91.7 percentage points**.

20.2 Future Work

- **Dense Vector Hybrid Retrieval:** Integrate local SQLite-VSS or ChromaDB vector embeddings alongside lexical BM25 for nuanced semantic search.
- **Multi-Modal Document Ingestion:** Extend MCP tools to parse PDF tables, lecture slides, and campus maps.
- **Distributed MCP Transports:** Implement Server-Sent Events (SSE) transports across networked Docker containers for distributed multi-server clusters.

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21. References

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